



CRYSTAL JACOBS

AI Automation & Data Engineer

DATA ENGINEERING · INTELLIGENT SYSTEMS · APPLIED AUTOMATION

Engineering intelligent data, automation and decision systems for real-world operational problems.

[GitHub](#)[LinkedIn](#)[Engineering Portfolio](#)[Credentials](#)

ENGINEERING PROFILE

AI Automation and Data Engineer with 15+ years of professional experience spanning data analytics, automation, systems integration, operational intelligence and technology-enabled business transformation. I design and implement systems that ingest, transform, analyse and operationalise data to support better decisions, reduce manual processes and improve operational visibility.

My engineering focus is connecting data, automation and intelligent systems to real-world operational problems. My background spans agriculture, manufacturing, medical/pharmaceutical, logistics and finance, giving me practical domain knowledge alongside a growing foundation in data engineering, AI and intelligent infrastructure.

TECHNICAL EXPERTISE

DATA ENGINEERING

- SQL & data modelling
- Data warehousing
- ETL / ELT pipelines
- Data quality & governance
- Analytics-ready architectures

ANALYTICS & BI

- Power BI
- Operational & financial analytics
- Executive reporting
- Data-driven decision support

AUTOMATION & INTEGRATION

- Workflow automation
- API integration
- ERP integration
- Process automation
- Systems integration

CLOUD & DATA PLATFORMS

- Azure
- SQL Server
- Databricks
- Apache Spark / PySpark
- Microsoft Fabric

AI & INTELLIGENT SYSTEMS

- Machine learning fundamentals
- Predictive modelling
- Intelligent decision support
- Applied AI / ML workflows

SOFTWARE & ENGINEERING

- Python
- SQL / T-SQL
- JavaScript
- Git / GitHub
- CI/CD foundations

Professional experience: Python, SQL, Power BI, Azure, SQL Server, ETL, automation and systems integration. Currently expanding: PySpark/Databricks, applied ML and electrical/electronic engineering foundations.

PROFESSIONAL EXPERIENCE

Senior Data Analyst and AI Automation Engineer

Bestbar Manufacturing | Bitberry IT (Westview Group) — Remote, Australian Hours · January 2024 – Present

Design, automate and deploy enterprise data systems supporting executive decision-making across Finance, Production, Manufacturing and Group Operations for Westview Group leadership.

CORE TECHNOLOGIES · SQL · Python · Power BI · Automation · APIs · ERP Systems · Database Management · Azure · Fabric

EXECUTIVE ANALYTICS & FINANCIAL INTELLIGENCE

- Delivered executive dashboards for Budget vs Actual, P&L, Forecasting, Labour Efficiency, Production KPIs and Operational Cost Analysis.
- Automated financial data ingestion from Business Central, Great Plains, Navision and SQL-based systems for real-time reporting.
- Built data models for MTD/YTD financial tracking, cash-flow trends and scenario modelling.

DATA ENGINEERING & AUTOMATION

- Engineered end-to-end ETL pipelines from ERP systems into SQL, on-premises and cloud environments.
- Delivered automated production dashboards used daily across factories for scheduling, material flow, throughput and QA monitoring.
- Eliminated manual reporting processes, cutting data availability time from hours to minutes.

AI / PREDICTIVE INTELLIGENCE

- Developed automation solutions using Python, TensorFlow, Pandas, PySpark and API integrations.
- Built predictive models for demand, production efficiency and inventory planning.

BUSINESS INTELLIGENCE & LEADERSHIP

- Designed enterprise reporting using Power BI, Python visualisation (Matplotlib/Plotly), SQL and custom APIs; maintained data integrity, governance and auditability across all layers.
- Team lead on priority automation initiatives; ran Agile sprints, defined scope, managed risk and coordinated across Australian and South African teams.

Senior Business Analyst (Agile Scrum)

Kaplan UK | Seaside Extended — Hybrid, United Kingdom Hours · June 2023 – December 2023

Delivered BA work inside a large-scale Microsoft Partner environment operating one of the largest custom Business Central development ecosystems in Africa.

- Led Scrum-based delivery cycles with developers, testers and product owners; produced requirements, process flows and end-to-end documentation for major modernisation initiatives.
- Contributed to Business Central custom development, Power Platform integrations and cloud-based data transformation using Microsoft Azure and analytical tooling.
- Supported the Solicitors Qualifying Examination (SQE) digital transformation, collaborating with Data Analysts, Product Managers and BAs to deliver Business Central enhancements.

Senior System Analyst & Developer

Afrilex Manufacturing (Pty) Ltd | Paarl, South Africa — Hybrid · February 2022 – May 2023

Pharmaceutical manufacturer specialising in medical innovation; led systems integration and clinical technology across the group.

SYSTEMS INTEGRATION & DEVELOPMENT

- Designed and implemented a full seed-to-sale pharmaceutical traceability system using Business Central AL, ensuring complete SAHPRA compliance and real-time tracking from cultivation to finished product.
- Engineered a custom ERP traceability platform aligned with medical/regulatory frameworks, improving auditing accuracy and operational reliability by over 50%.
- Managed vendor and customer relationships across the integration lifecycle.

CLINICAL TECHNOLOGY DEVELOPMENT

- Built a clinical research traceability platform (HTML, CSS, JavaScript, Azure API integrations) supporting medical research, documentation automation and compliance.
- Automated clinical trial document workflows, reducing manual effort and turnaround by 40–60%.
- Developed secure SharePoint hubs across multiple companies for compliance documentation, roles and approval flows; rolled out Microsoft 365 organisation-wide; used SQL for regulatory data analysis and validation.

PROMOTION — AGRONOMY PRODUCTION MANAGER

- Led digital transformation of facility operations at a medically registered cannabis facility: IoT systems, HVAC automation and environmental data monitoring for compliance, energy efficiency and production stability.
- Enabled data-driven decision-making across cultivation and processing.

Chief Executive Officer

TauMililo Transport Services — South African Time Zone · December 2014 – November 2020

National and cross-border logistics company (Agricultural, FMCG and speciality freight, warehousing, temperature-sensitive transport).

- Led the full operational ecosystem: transport, warehousing, client delivery and financial planning across a multi-region network.
- Used fleet utilisation analytics, cost modelling and route optimisation — SQL-based reporting and telematics dashboards — to improve margins, reduce downtime and improve fuel efficiency.
- Built and monitored operational datasets (fleet utilisation, delivery performance, warehouse throughput) to support forecasting and executive decision-making.
- Managed regulatory compliance (ISO 9001 warehouse standards, SADC cross-border freight); oversaw workforce planning, safety and operational KPIs.
- Led contract negotiations and service-level governance for enterprise agriculture and FMCG clients.

Operations and Livestock Manager

Conquela Agrilogistics — On-Farm & Travel, South Africa · January 2008 – October 2014

Full-scale agricultural operations: feedlots, small-stock herd management, crop production, stud genetic breeding on global bloodlines.

- Directed feedlot, dairy and small-stock operations — irrigation systems, crop cycles, nutrition planning and stud breeding programmes (A.I. and embryo transfer) — for productivity, sustainability and regulatory compliance.
- Oversaw crop production, soil management and irrigation optimisation, improving yield and resource efficiency through field monitoring and research-based practice.
- Implemented structured data tracking for feedlot performance, crop yields and resource usage to support forecasting and decision-making.
- Managed auction preparation, livestock sales and genetics marketing for high-value stud bloodlines.

This role is foundational to my engineering identity — it built the operational, data and forecasting instincts I now apply to software and automation systems.

SELECTED ENGINEERING PROJECTS

A selection of engineering projects that demonstrate my expertise in data engineering, automation, AI systems and intelligent solutions.



Data Warehouse Project

LIVE

Production-style SQL Server data warehouse: end-to-end ingestion, ETL, data quality enforcement, dimensional modelling and analytics-ready reporting.

SQL Server

T-SQL

Star Schema

ETL

View on GitHub →



Professional Engineering Portfolio

LIVE

Responsive engineering portfolio built as a technical system for presenting engineering work, credentials and capabilities.

HTML/CSS/JS

Git

GitHub Pages

Responsive

View on GitHub →



Vitaray Athena

IN DEVELOPMENT

An evolving intelligence platform focused on environmental, resource and industrial risk intelligence and decision support.

Python

SQL

AI/ML

APIs

Vector DB



Smart Agriculture System

IN DEVELOPMENT

Agricultural intelligence system combining data, automation and IoT-oriented technology for resource management and decision-making.

Python

SQL

IoT

AI/ML

APIs



Financial Tracking System

PLANNED

Python-based financial tracking and analytical system: structured data processing, financial analysis, automation and reporting.

Python

Pandas

NumPy

SQL

Power BI



Energy Grid Intelligence

PLANNED

Intelligent energy system for monitoring, analytics, forecasting, anomaly detection and optimisation.

Python

SQL

Time Series

AI/ML

IoT

More projects & code: Explore my repositories and engineering work on [GitHub](#) and my [Engineering Portfolio](#).

EDUCATION



Bachelor of Engineering Technology in Electrical and Electronic Engineering (NQF7)

PLANNED

Planned — February 2027



Bachelor of Science in Information Technology — Software Engineering

Richfield University · February 2019 – December 2021 · Final Year GPA: 3.8 / 10 (Distinctions)

ADDITIONAL AGRICULTURAL & TECHNICAL EDUCATION

- S.A.C. Diploma in Agricultural Management**
Stonebridge College · 2009–2011 · GPA 3.8/90%+
- Advanced Agronomy & Plant Nutrition**
Nosa Agriculture · 2006–2007

- **Computer Science Diploma**
John's Academy · 2018–2019
 - **Stable Management**
Stonebridge College · 2008–2010
 - **Wool Classification**
Cape Wool & Mohair · 2013
 - **A.I. & Pregnancy Practical Accreditation**
Rothman Livestock Services · 2010
 - **Sheep Classification**
Dorper Sheep Breeders Society · 2009–2010
- **Sheep Husbandry Advanced Certificate**
ACS Distance Education · 2012–2013
 - **Sustainable Agriculture**
Nutri-Tech Solutions · 2013
 - **Stockman Certification**
Aldam Stockman's School · 2012
 - **Livestock Management**
Mildex Livestock Training · 2010

Additional credentials available via Engineering Portfolio →

PROFESSIONAL CERTIFICATIONS



GitHub Foundations ACTIVE

Achieved August 2026

[View Credential →](#)

GH-900



PCEP — Certified Entry-Level Python Programmer ACTIVE

Achieved 8 September 2025

[View Credential →](#)

PCEP-30-02



Microsoft Power Platform Functional Consultant Associate ACTIVE

Achieved 8 May 2024

[View Credential →](#)

PL-200



Microsoft Business Central Functional Consultant Associate ACTIVE

Achieved 25 September 2023

[View Credential →](#)

MB-800

Additional credentials available via Engineering Portfolio →

PUBLICATIONS & INDUSTRY RECOGNITION

- 

2011 — **Hereford Journal** | Top Breeder & Biography
- 

2013 — **Landbouweekblad** | Aldam Stockman's School, Top 10 Achievement
- 

2014 — **Hereford Youth Show** | Championship Recognition

ENGINEERING DIRECTION

Building toward intelligent infrastructure by connecting data engineering, automation and AI with sensors, software and physical systems across agriculture, energy and industrial environments.

**CURRENT FOUNDATION**

Data Engineering
AI Automation
Intelligent Systems

**ENGINEERING EXPANSION**

Electrical & Electronic Engineering
IoT
Embedded Systems

**APPLIED SYSTEMS**

Agriculture
Energy
Intelligent Infrastructure

INDUSTRY & DOMAIN EXPERIENCE

 Agriculture

 Manufacturing

 Finance

 Medical

 Logistics

 Intelligent Infrastructure

Confidential — provided for recruitment purposes only. Unauthorised reproduction or distribution is prohibited.